# Periodic and Oscillatory Motion

*Class II · $x$ measured from EQUILIBRIUM · ideal MASSLESS springs, FRICTIONLESS surfaces, uniform $g$ unless stated · $\omega$ = angular frequency of the undamped SHM, $\omega_d$ = driving*

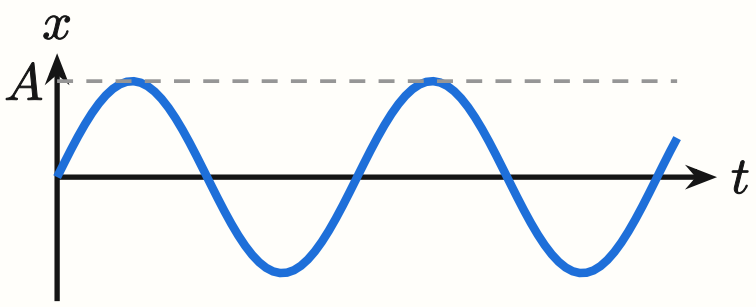
## 1 Periodic motion

$$T = \frac{1}{f} \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

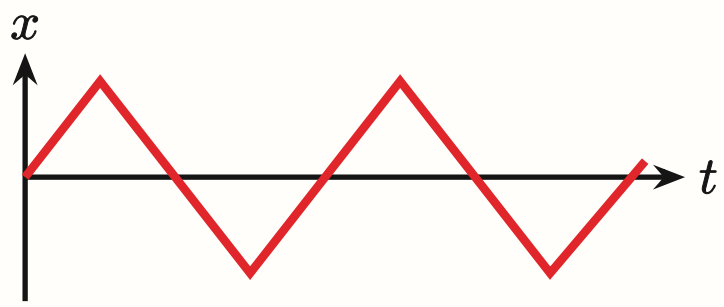
*REPEATS after a fixed time $T$ (s) · $f$ in Hz · $\omega$ in $\mathrm{rad\,s^{-1}}$*

- *Oscillatory: to and fro about an EQUILIBRIUM position. Every oscillation is periodic; uniform circular motion is periodic, not oscillatory.*

## 2 Oscillatory, but is it SHM?




*SHM: $x = A\sin\omega t$*




*periodic, oscillatory, NOT SHM*

## 3 Which functions?

- $\sin\omega t + \cos\omega t = \sqrt{2}\sin(\omega t + \frac{\pi}{4})$: *SHM, amplitude $\sqrt{2}$, period $\frac{2\pi}{\omega}$.*
- $\sin^2\omega t = \frac{1}{2} - \frac{1}{2}\cos 2\omega t$: *SHM about $x = \frac{1}{2}$, period $\frac{\pi}{\omega}$.*
- $\sin\omega t + \sin 2\omega t$: *periodic (period $\frac{2\pi}{\omega}$) but NOT SHM. $e^{-\omega t}$: not periodic.*

# Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

## 1 JEE Main + JEE Advanced + NEET syllabus

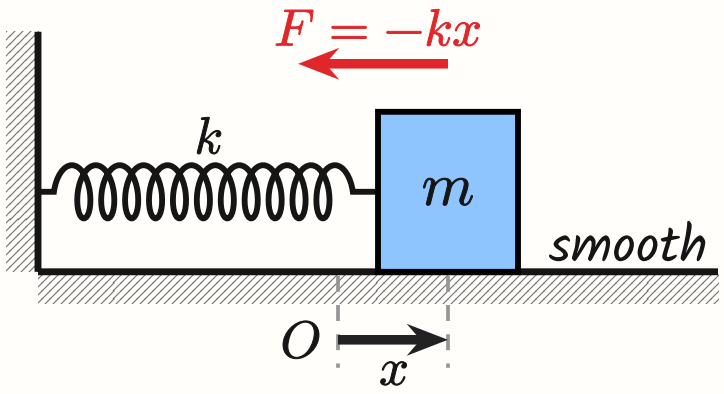
- ✓ Periodic motion: period, frequency — p1
- ✓ SHM: $F = -kx$, $a = -\omega^2 x$; equations, phase — p3, p4
- ✓ Phase relations, $x$, $v$, $a$ graphs; $v$–$x$, $a$–$x$ — p5–p7
- ✓ Energy in SHM: KE, PE, total; graphs; averages — p8, p9
- ✓ Spring–mass: horizontal, vertical, series, parallel — p10, p11
- ✓ Spring cut into parts; massive spring correction — p12
- ✓ Simple pendulum (small angle); $g_{eff}$ in a lift, cart, fluid — p13, p14, p16
- ✓ Physical (compound) and torsional pendulum — p17, p18
- ✓ Damped oscillations; forced oscillations, resonance — p25–p27
- ✓ Pendulum tension; seconds pendulum; large amplitude — p13

## 2 Also covered: systems + standard results

- ✓ Reference circle; time between two positions — p6
- ✓ Seconds pendulum; very long pendulum ($l \sim R$) — p13, p15
- ✓ U-tube liquid; floating body; gas column, piston — p19, p20
- ✓ Tunnel through the earth — p21
- ✓ SHM from $U(x)$: $\omega^2 = U''(x_0)/m$ — p22
- ✓ Superposition: same direction, perpendicular — p23, p24
- ✓ Two bodies on a spring (reduced mass) — p12
- ✓ Angular SHM: restoring torque $\tau = -\kappa\theta$ — p3, p17, p18
- ✓ Special cases, results, traps, solved questions — p28–p37

# SHM: Definition

## 1 Restoring force




$$F = -kx \qquad a = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

$x$ from EQUILIBRIUM $O$ · $k$ = force constant ($\mathrm{N\,m^{-1}}$) · ideal massless spring, frictionless surface

## 2 Definition

- SHM: acceleration (or restoring force) is proportional to the displacement from equilibrium and always directed TOWARDS it.
- Equation of motion: $\dfrac{d^2x}{dt^2} + \omega^2 x = 0$. Only a LINEAR restoring force gives SHM; $F = -kx^3$ oscillates, but it is not SHM.
- Angular SHM: $\tau = -\kappa\theta$ gives $\alpha = -\omega^2\theta$ with $\omega^2 = \kappa/I$.

## 3 Reading $\omega$ off an equation

**Example**

$a = -16x$ (SI units): $\omega^2 = 16$, $\omega = 4\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{4} = 1.57\,\mathrm{s}$.

$a = -16x + 8$: SHM about $x_0 = 0.5\,\mathrm{m}$ (where $a = 0$), same $\omega = 4\,\mathrm{rad\,s^{-1}}$.

# Equations of SHM

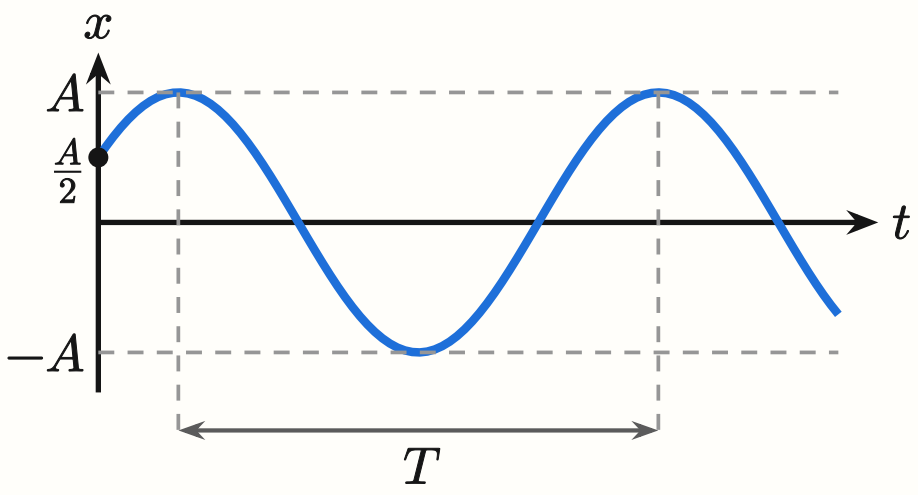
## 1 Displacement, velocity, acceleration

$$x = A\sin(\omega t + \phi)$$

$$v = A\omega\cos(\omega t + \phi) \qquad a = -A\omega^2\sin(\omega t + \phi)$$

$a = -\omega^2 x$ · $x$ from equilibrium · $A$ = amplitude (maximum $|x|$) · $(\omega t + \phi)$ = PHASE (rad), fixes position AND direction · $\phi$ = initial phase

## 2 The graph




- Drawn: $\phi = \frac{\pi}{6}$, so $x(0) = \frac{A}{2}$, moving in $+x$.
- $\phi = 0$: at $O$, moving $+x$. $\phi = \frac{\pi}{2}$: at $+A$. $\phi = \pi$: at $O$, moving $-x$.

## 3 Amplitude and $\phi$ from the start

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \qquad \tan\phi = \frac{\omega x_0}{v_0}$$

$x_0$, $v_0$ at $t = 0$ · for $x = A\sin(\omega t + \phi)$ · choose the quadrant from the SIGNS: $\sin\phi = \frac{x_0}{A}$, $\cos\phi = \frac{v_0}{A\omega}$

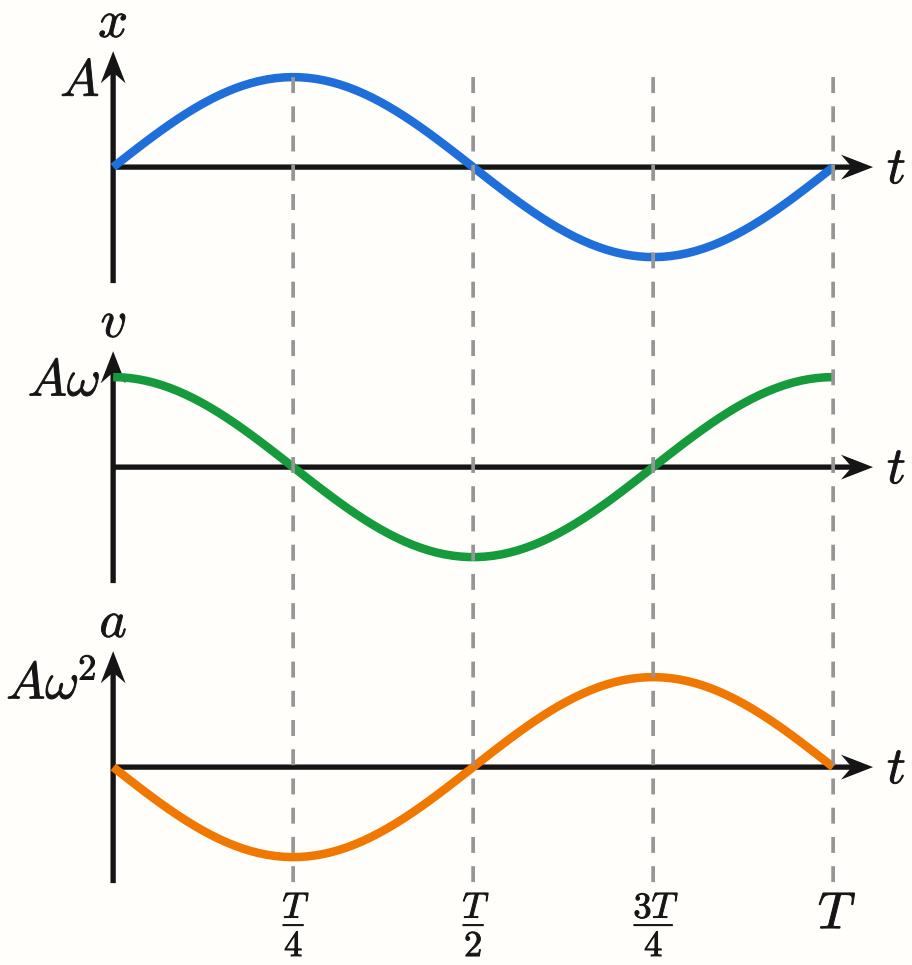
**Example**

$\omega = 2\,\text{rad}\,\text{s}^{-1}$, $x_0 = 3\,\text{cm}$, $v_0 = 8\,\text{cm}\,\text{s}^{-1}$: $A = \sqrt{9 + 16} = 5\,\text{cm}$; $\sin\phi = 0.6$, $\cos\phi = 0.8$, $\phi = 37°$.

# Phase Relations and Graphs

$x = A\sin\omega t$ (initial phase zero) for the graphs

## 1 $x$, $v$, $a$ against $t$




$v$ leads $x$ by $\frac{\pi}{2}$

$a$ leads $v$ by $\frac{\pi}{2}$

$a$ and $x$ : phase $\pi$

same $\omega$ for all three · $a = -\omega^2 x$ always

## 2 Reading the graphs

- At the MEAN position: $|v| = A\omega$ (max), $a = 0$. At the EXTREMES: $v = 0$, $|a| = A\omega^2$ (max).
- Slope of the $x$–$t$ graph $= v$; slope of the $v$–$t$ graph $= a$.

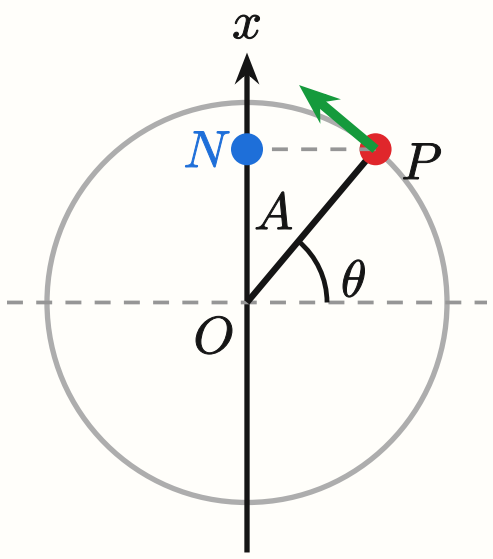
## 3 Phase difference of two particles

**Example**

Two particles, same $A$ and $\omega$, cross at $x = \frac{A}{2}$ moving in OPPOSITE directions: phases $\frac{\pi}{6}$ and $\frac{5\pi}{6}$, phase difference $\frac{2\pi}{3}$.

# Reference Circle and Time

## 1 SHM = projection of uniform circular motion




$$ON = A\sin\theta \qquad \theta = \omega t + \phi$$

*P: circle of radius $A$, CONSTANT $\omega$ · its projection $N$ on a diameter performs SHM*

## 2 Time between positions

$$0 \to \frac{A}{2} : \frac{T}{12} \qquad \frac{A}{2} \to A : \frac{T}{6}$$

$$0 \to \frac{A}{\sqrt{2}} : \frac{T}{8} \qquad 0 \to \frac{\sqrt{3}A}{2} : \frac{T}{6}$$

*along the direct path, no turning back · $\Delta t = \Delta(\text{phase})/\omega$*

**Example**

$T = 2.4\,\text{s}$, $x = \frac{A}{2}$ *through* $O$ *to* $-\frac{A}{2}$: $2 \times \frac{T}{12} = 0.4\,\text{s}$.

## 3 Distance

- *Distance:* $4A$ *per period,* $2A$ *per half period from ANY start;* $A$ *in* $\frac{T}{4}$ *only from* $O$ *or an extreme.*
- *Average speed over a period* $= \frac{4A}{T} = \frac{2A\omega}{\pi}$; *average velocity over a period* $= 0$.

# Velocity and Acceleration vs $x$

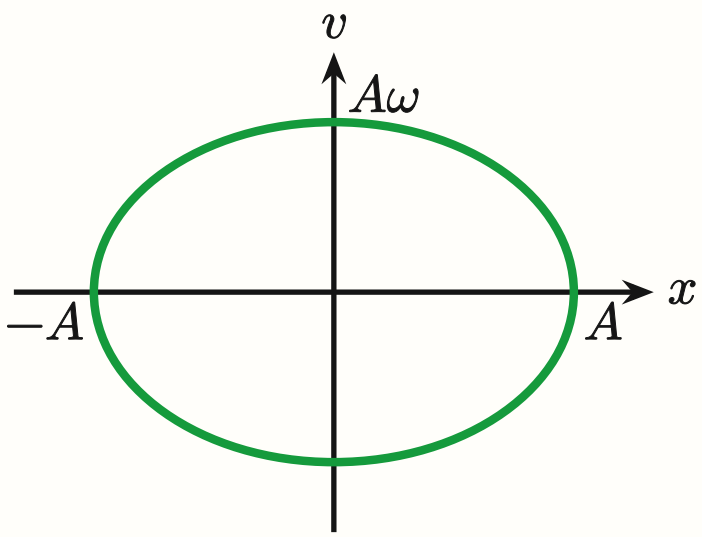
## 1 $v$ and $a$ in terms of $x$

$$v = \pm\omega\sqrt{A^2 - x^2} \qquad a = -\omega^2 x$$

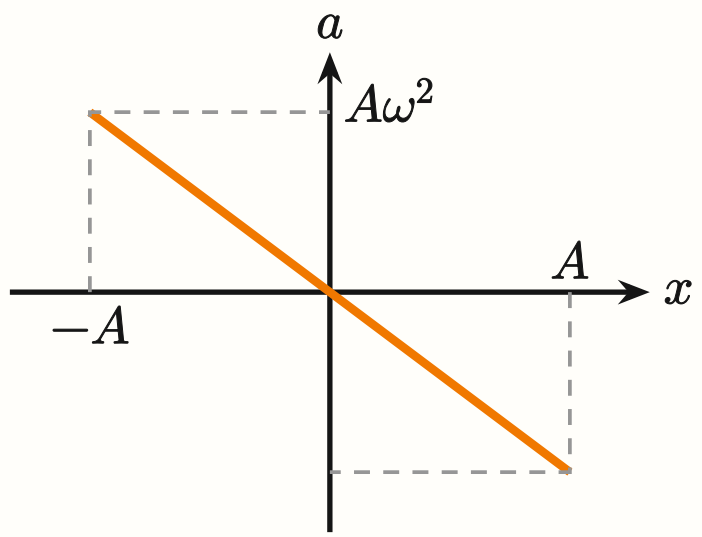
$$v_{max} = A\omega \quad (x = 0) \qquad a_{max} = A\omega^2 \quad (x = \pm A)$$

*$x$ from equilibrium · sign of $v$ = direction of motion · from $v^2 = \omega^2(A^2 - x^2)$*

## 2 Graphs




ellipse $\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1$




straight line, slope $-\omega^2$

## 3 Speeds at two positions

$$\omega^2 = \frac{v_1^2 - v_2^2}{x_2^2 - x_1^2} \qquad A^2 = \frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}$$

*speed $v_1$ at $x_1$, speed $v_2$ at $x_2$, same SHM*

**Example**

$4\,\mathrm{cm\,s^{-1}}$ *at* $3\,\mathrm{cm}$, $3\,\mathrm{cm\,s^{-1}}$ *at* $4\,\mathrm{cm}$: $\omega^2 = \frac{16-9}{16-9} = 1$, $\omega = 1\,\mathrm{rad\,s^{-1}}$;
$A^2 = \frac{256-81}{7} = 25$, $A = 5\,\mathrm{cm}$.

# Energy in SHM

*PE taken ZERO at the equilibrium position · no friction, no damping*

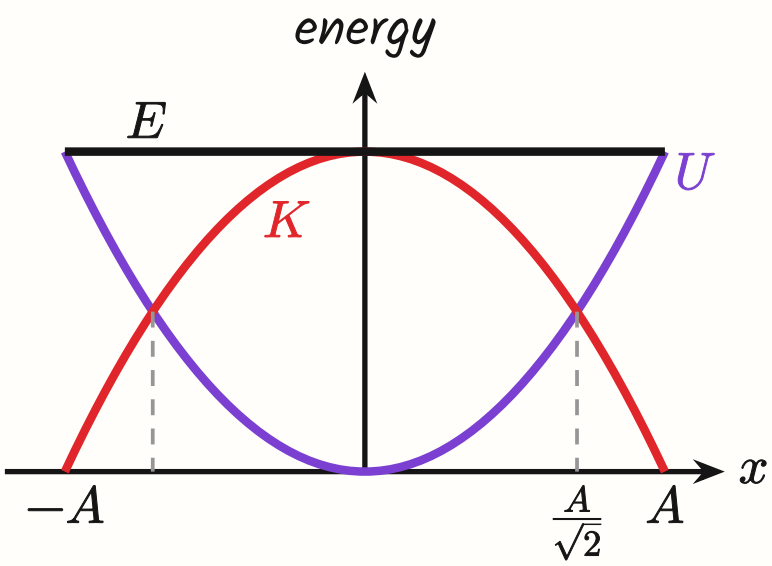
## 1 Kinetic, potential, total

$$K = \tfrac{1}{2}m\omega^2(A^2 - x^2) \qquad U = \tfrac{1}{2}m\omega^2 x^2$$

$$E = K + U = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$k = m\omega^2$ · $E$ CONSTANT (no friction, no damping) · $E \propto A^2$ and $\propto \omega^2$

## 2 Energy against $x$




- Parabolas: $U$ max at the extremes, $K$ max at $O$.
- $K = U$ at $x = \pm\frac{A}{\sqrt{2}}$.
- At $x = \frac{A}{2}$: $K = \frac{3}{4}E$, $U = \frac{1}{4}E$.

## 3 Numbers; vertical spring

**Example**

$m = 0.2\,\text{kg}$, $k = 20\,\text{N m}^{-1}$, $A = 0.1\,\text{m}$: $E = \frac{1}{2} \times 20 \times 0.01 = 0.1\,\text{J}$.
At $x = 0.05\,\text{m}$: $U = 0.025\,\text{J}$, $K = 0.075\,\text{J}$;
$v_{max} = \sqrt{2E/m} = 1\,\text{m s}^{-1}$.

- Vertical spring: measure $x$ from the NEW equilibrium; spring PE + gravity PE together then equal $\frac{1}{2}kx^2 +$ constant, so the same formulas hold.

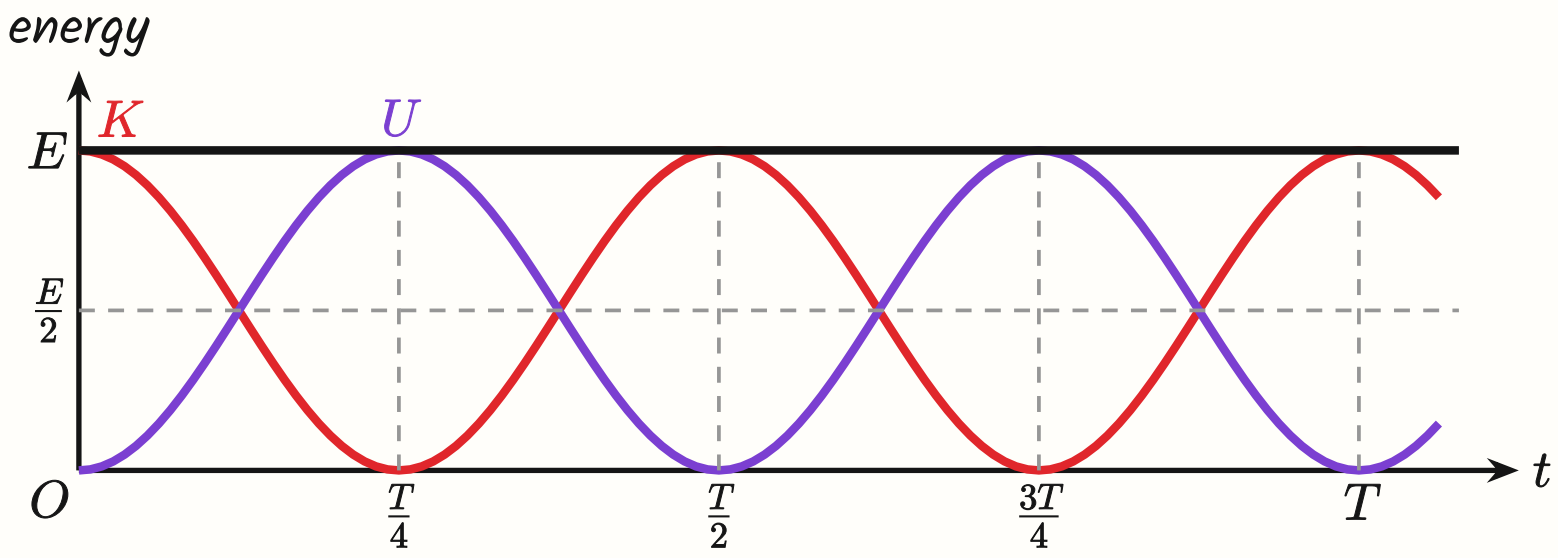
# Energy vs Time; Averages

$x = A\sin\omega t$ · **PE zero at equilibrium · no damping**

## 1 Energy against $t$

$$K = E\cos^2\omega t \qquad U = E\sin^2\omega t$$

$E = \frac{1}{2}m\omega^2A^2$ · $K$ and $U$ oscillate with period $\frac{T}{2}$ (frequency $2f$) between $0$ and $E$




## 2 Average values

$$\langle K\rangle_t = \langle U\rangle_t = \frac{E}{2}$$

$$\langle K\rangle_x = \frac{2}{3}E \qquad \langle U\rangle_x = \frac{1}{3}E$$

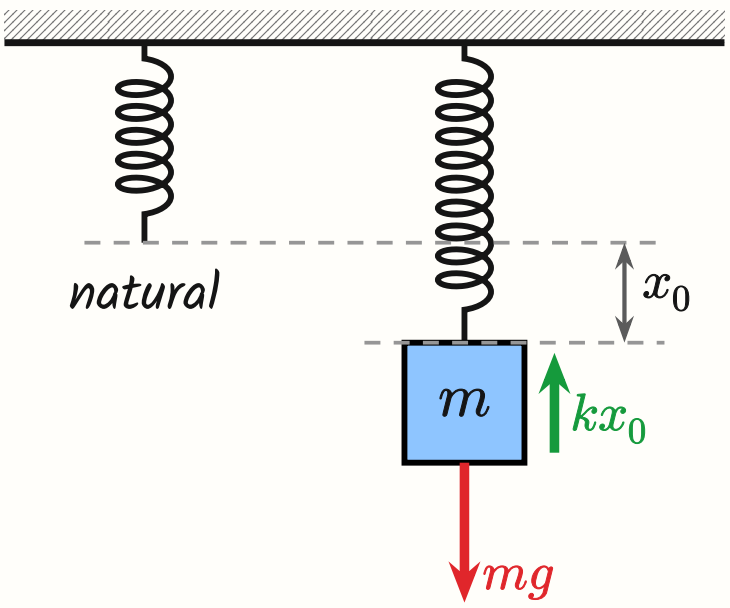
$\langle\ \rangle_t$: averaged over one period in TIME · $\langle\ \rangle_x$: averaged over POSITION, $x$ from $-A$ to $A$

- $K = U$ first at $t = \frac{T}{8}$ (phase $\frac{\pi}{4}$, $x = \frac{A}{\sqrt{2}}$), then every $\frac{T}{4}$.
- The particle spends MORE time near the extremes (it is slow there), so the time average of $U$ exceeds the position average of $U$.

# Spring-Block Systems

*ideal massless spring · rigid support · $x$ from the equilibrium position*

## 1 Horizontal and vertical




$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$x_0 = \frac{mg}{k} \qquad T = 2\pi\sqrt{\frac{x_0}{g}}$$

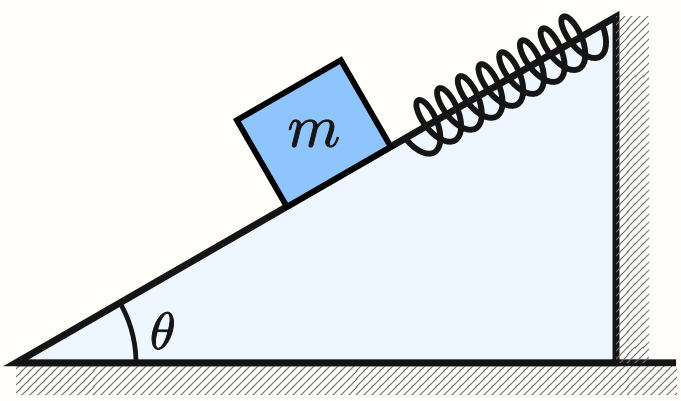
*horizontal (smooth floor) OR vertical: SAME $T$ · vertical: SHM about the stretched equilibrium, $x_0$ = static extension · $T$ independent of $A$*

**Example**

*Static extension* $x_0 = 9.8\,\text{cm}$, $g = 9.8\,\text{m s}^{-2}$:

$T = 2\pi\sqrt{0.098/9.8} = 2\pi \times 0.1 = 0.628\,\text{s}.$

## 2 On a smooth incline




$$T = 2\pi\sqrt{\frac{m}{k}}$$

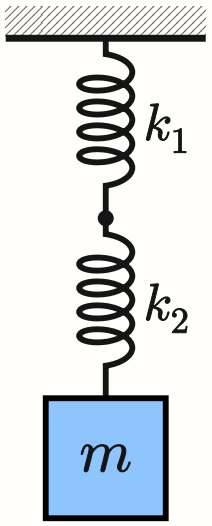
$$x_0 = \frac{mg\sin\theta}{k}$$

*smooth incline, spring along the slope, fixed at the top · gravity only SHIFTS the equilibrium, never changes $T$*

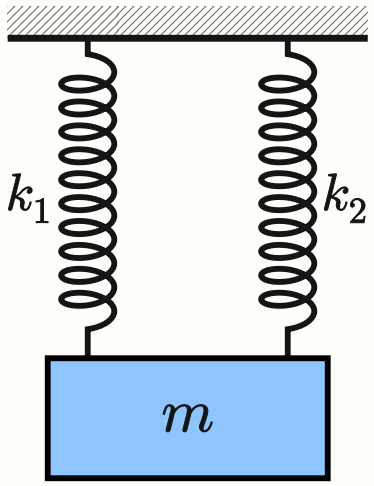
# Springs in Series and Parallel

*ideal massless springs · smooth floor · rigid supports*

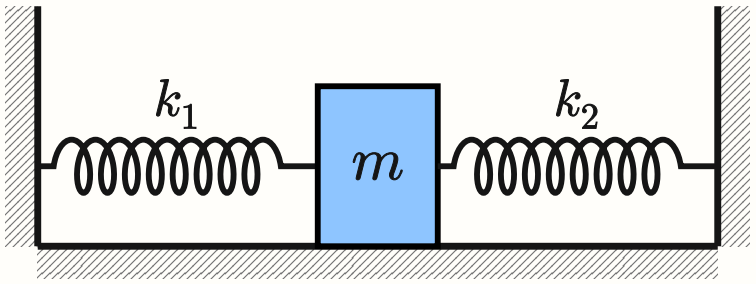
## 1 The three arrangements




series




parallel




between two walls

## 2 Effective spring constant

series: $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$ parallel: $k = k_1 + k_2$

block between two walls: $k = k_1 + k_2$

*series: SAME force, extensions add · parallel: SAME extension, forces add · between walls: both springs pull the block back the same way: PARALLEL*

- Series of two equal springs: $\frac{k}{2}$, $T$ grows $\sqrt{2}$ times. Parallel: $2k$, $T$ falls $\sqrt{2}$ times.

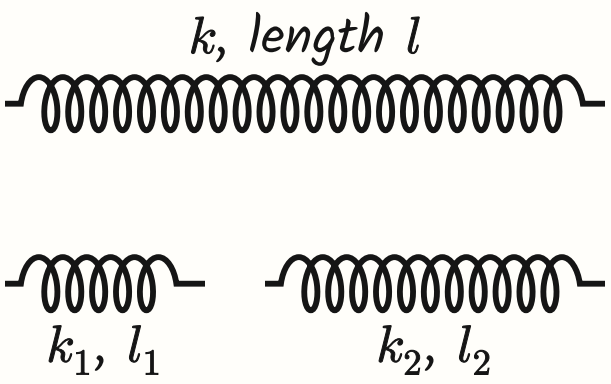
**Example**

$k_1 = 200\,\mathrm{N\,m^{-1}}$, $k_2 = 300\,\mathrm{N\,m^{-1}}$, $m = 3\,\mathrm{kg}$. Series: $k = 120\,\mathrm{N\,m^{-1}}$, $T = 2\pi\sqrt{3/120} = 0.993\,\mathrm{s}$. Parallel: $k = 500\,\mathrm{N\,m^{-1}}$, $T = 2\pi\sqrt{3/500} = 0.487\,\mathrm{s}$.

# Cut Springs, Two Bodies, Massive Spring

*uniform spring · smooth floor · small oscillations*

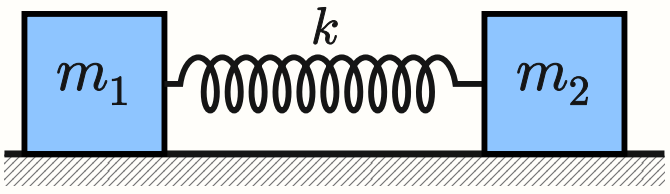
## 1 Spring cut into parts




$$k\,l = k_1 l_1 = k_2 l_2$$

*UNIFORM spring:* $k \propto \frac{1}{l}$ · *n equal parts: each* $nk$; *all* $n$ *in parallel:* $n^2 k$ · $k = 100\,\mathrm{N\,m^{-1}}$ *cut* $1:2$: $300$ *and* $150\,\mathrm{N\,m^{-1}}$

## 2 Two blocks joined by a spring




$$T = 2\pi\sqrt{\frac{\mu}{k}} \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

*no external horizontal force (centre of mass at rest)* · $\mu$ = *reduced mass* · $m_1 = m_2 = 2\,\mathrm{kg}$, $k = 100\,\mathrm{N\,m^{-1}}$: $\mu = 1\,\mathrm{kg}$, $T = 0.628\,\mathrm{s}$

## 3 Spring with mass

$$T = 2\pi\sqrt{\frac{m + m_s/3}{k}}$$

*UNIFORM spring of mass* $m_s$, *one end FIXED, block* $m$ *at the other* · *speed of the coils grows linearly along the spring*

**Example**

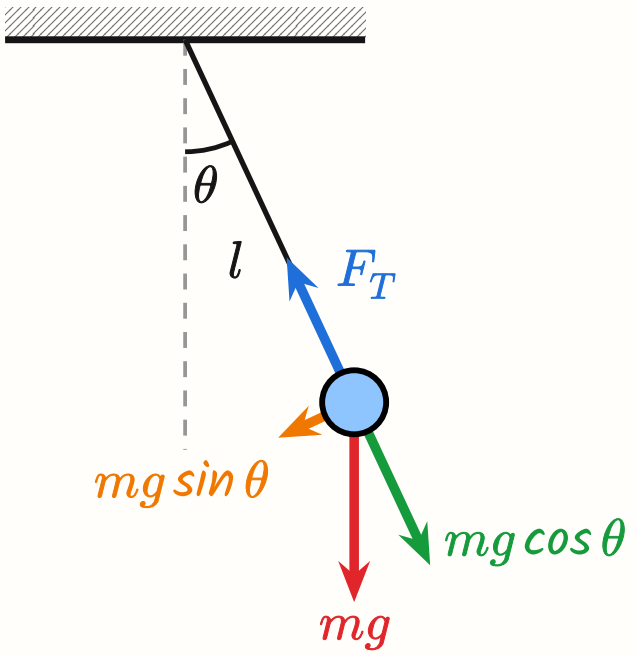
$m = 1\,\mathrm{kg}$, $m_s = 0.3\,\mathrm{kg}$, $k = 110\,\mathrm{N\,m^{-1}}$:

$T = 2\pi\sqrt{1.1/110} = 0.628\,\mathrm{s}.$

# Simple Pendulum

*point bob, light inextensible string, rigid support, uniform $g$, no air drag*

## 1 Forces on the bob




$$T = 2\pi\sqrt{\frac{l}{g}}$$

SMALL angle ($\theta_0 \lesssim 10°$, $\sin\theta \approx \theta$) · $l$ = support to the bob's CENTRE · independent of $m$ and of a small amplitude

- Restoring force $mg\sin\theta \approx \frac{mg}{l}s$ ($s$ = arc): linear only for small $\theta$.
- Tension: $F_T = mg\cos\theta + \frac{mv^2}{l}$; maximum at the mean position, $mg(3 - 2\cos\theta_0)$.

**Example**

$l = 1\,\text{m}$, $g = 9.8\,\text{m}\,\text{s}^{-2}$, small angle: $T = 2\pi\sqrt{1/9.8} = 2.007\,\text{s}$.

## 2 Seconds pendulum and large amplitude

$$T = 2\,\text{s} \;\Rightarrow\; l = \frac{g}{\pi^2} = 0.993\,\text{m}$$

$$T \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\theta_0^2}{16}\right)$$

$g = 9.8\,\text{m}\,\text{s}^{-2}$ · second line: moderate amplitude $\theta_0$ (rad), first correction only

# Pendulum in Accelerating Frames

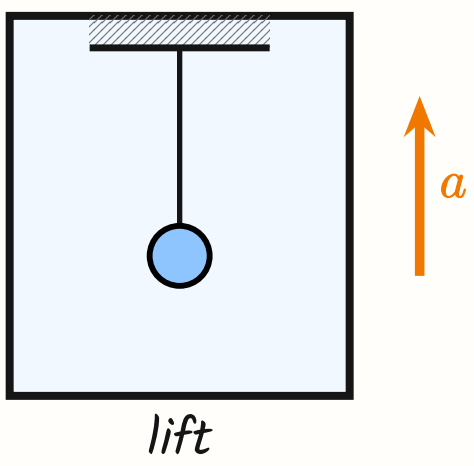
*small angle · $\vec{a}$ = acceleration of the support (lift / cart) · uniform $g$*

## 1 Effective $g$

$$T = 2\pi\sqrt{\frac{l}{g_{eff}}} \qquad \vec{g}_{eff} = \vec{g} - \vec{a}$$

*pseudo force $-m\vec{a}$ in the support's frame · the mean position lies along $\vec{g}_{eff}$*

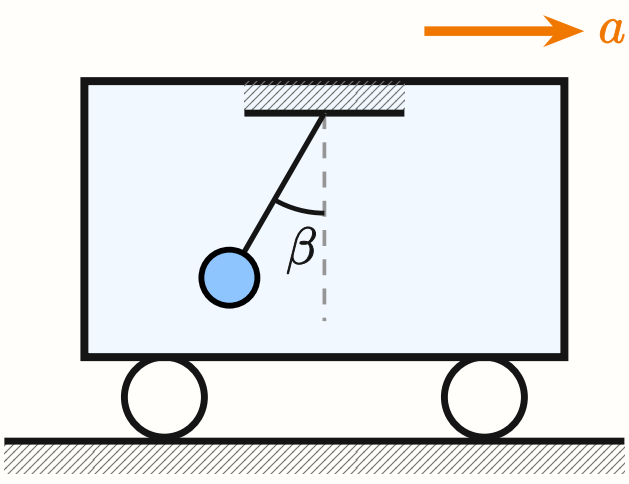
## 2 Lift




*lift*

- *Accelerating UP:* $g_{eff} = g + a$, $T$ *smaller.*
- *Accelerating DOWN* ($a < g$): $g_{eff} = g - a$, $T$ *larger. Free fall: no oscillation. Constant velocity:* $T$ *unchanged.*
- $T_0 = 2\,\mathrm{s}$, $a = \frac{g}{4}$ *up:* $T = \frac{2}{\sqrt{1.25}} = 1.789\,\mathrm{s}$.

## 3 Cart with horizontal acceleration




$$g_{eff} = \sqrt{g^2 + a^2}$$

$$\tan\beta = \frac{a}{g}$$

*the string hangs tilted BACK (opposite to $\vec{a}$) by $\beta$ and oscillates about that line*

# Very Long Pendulum

*small angle · light string, rigid support · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ at the surface*

## 1 Length comparable to the earth's radius

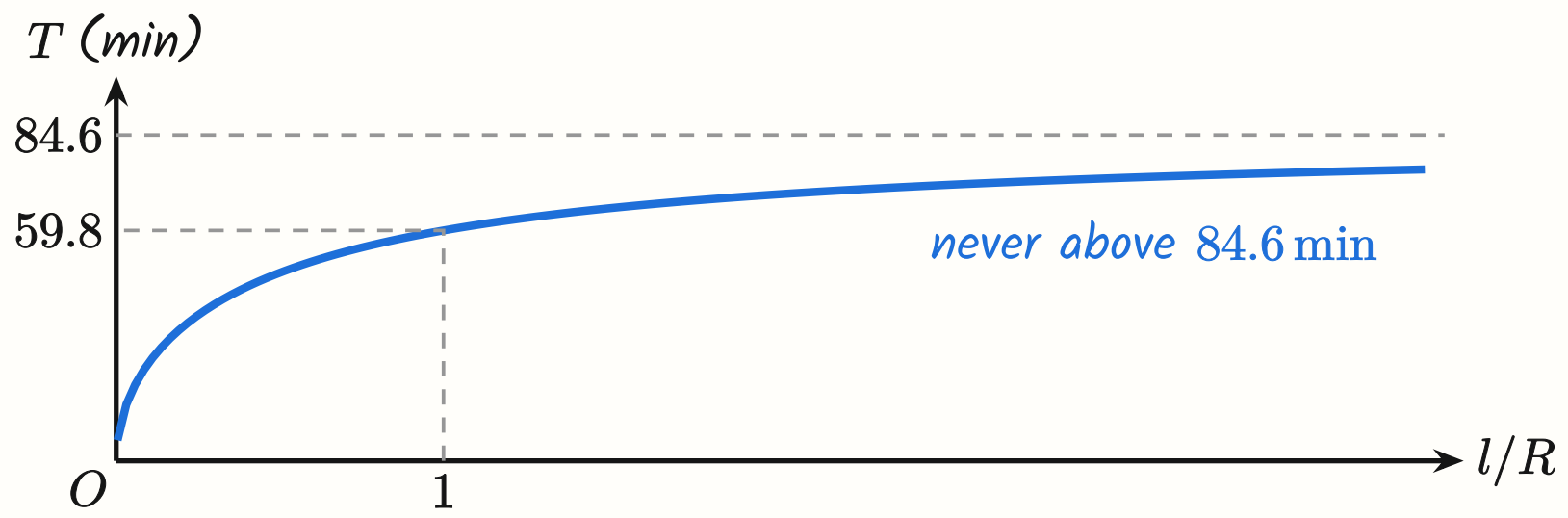
$$T = 2\pi\sqrt{\frac{1}{g\left(\frac{1}{l}+\frac{1}{R}\right)}}$$

$$l \to \infty:\quad T = 2\pi\sqrt{\frac{R}{g}} = 84.6\,\mathrm{min}$$

*bob near the surface, $g$ directed to the earth's CENTRE (not parallel) · $l \ll R$ gives back $2\pi\sqrt{l/g}$*

- $l = R$: $T = 2\pi\sqrt{\frac{R}{2g}} = 59.8\,\mathrm{min}$.
- *Why: for a long string the restoring force has TWO parts, from the string's tilt ($\propto \frac{s}{l}$) and from $\vec{g}$ turning towards the centre ($\propto \frac{s}{R}$), $s$ = arc moved by the bob.*

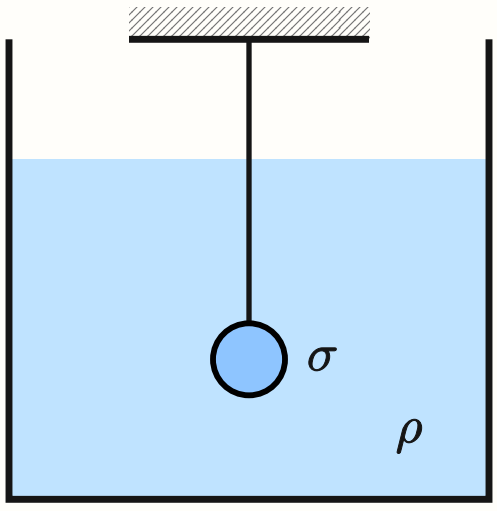
## 2 Period against length



# Pendulum in a Fluid or a Field

*small angle · rigid support · uniform* $g$ · $T_0 = 2\pi\sqrt{l/g}$

## 1 Bob immersed in a liquid




$$g_{eff} = g\left(1 - \frac{\rho}{\sigma}\right)$$

*bob density* $\sigma$, *liquid density* $\rho < \sigma$ · *buoyancy only: NON-viscous liquid, drag neglected*

**Example**

$\sigma = 4\rho$: $g_{eff} = \frac{3}{4}g$, $T = \frac{T_0}{\sqrt{0.75}} = 1.155\,T_0$.

## 2 Charged bob in a uniform field

$$g_{eff} = g + \frac{qE}{m} \quad (q\vec{E} \text{ down})$$

$$g_{eff} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \quad (\vec{E} \text{ horizontal})$$

*uniform* $\vec{E}$, *charge* $q$ *on a bob of mass* $m$, *insulating string* · $q\vec{E}$ *up:* $g - \frac{qE}{m}$

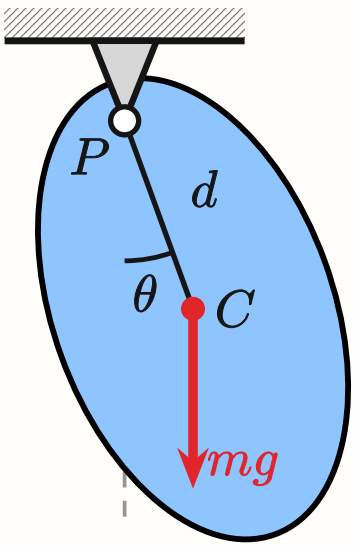
**Example**

$m = 10\,\text{g}$, $q = 10\,\mu\text{C}$, $\vec{E}$ *downward*, $E = 4900\,\text{N C}^{-1}$:
$\frac{qE}{m} = 4.9\,\text{m s}^{-2}$, $g_{eff} = 14.7\,\text{m s}^{-2}$ $(g = 9.8\,\text{m s}^{-2})$,
$T = \frac{T_0}{\sqrt{1.5}} = 0.816\,T_0$.

# Physical (Compound) Pendulum

*rigid body · fixed HORIZONTAL axis through the pivot, frictionless · small angle · uniform $g$*

## 1 Rigid body about a pivot




$$T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{k^2 + d^2}{gd}}$$

*$I$ about the PIVOT $= m(k^2 + d^2)$ · $k$: radius of gyration about $C$ · $d = PC$*

- *Equivalent length $d + \frac{k^2}{d}$; least $T$ at $d = k$: $T_{min} = 2\pi\sqrt{\frac{2k}{g}}$.*

## 2 Standard bodies

*rod, end:* $2\pi\sqrt{\frac{2L}{3g}}$ *ring on a nail:* $2\pi\sqrt{\frac{2R}{g}}$

*disc, rim point:* $2\pi\sqrt{\frac{3R}{2g}}$

*uniform rod, length $L$ · ring / disc, radius $R$, axis $\perp$ its plane*

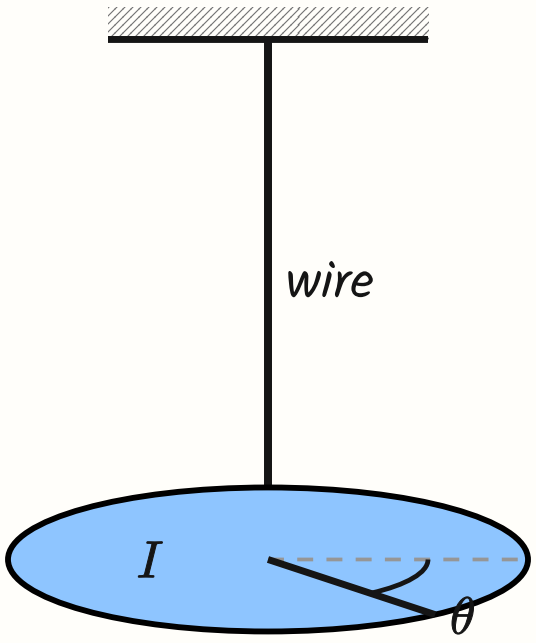
**Example**

*Rod, $L = 1.5\,\mathrm{m}$, end pivot, $g = 9.8\,\mathrm{m\,s^{-2}}$:* $T = 2\pi\sqrt{\frac{2\times1.5}{3\times9.8}} = 2.007\,\mathrm{s}$ *(as a $1\,\mathrm{m}$ simple pendulum).*

# Torsional Pendulum

*wire within the elastic limit · wire mass neglected · clamped rigid support*

## 1 Twisted wire




$$\tau = -C\theta \qquad T = 2\pi\sqrt{\frac{I}{C}}$$

$C$ = torsion constant ($\mathrm{N\,m\,rad^{-1}}$) · $I$ about the WIRE axis · linear for any twist within the elastic limit (no small-angle condition)

$$C = \frac{\pi\eta r^4}{2l}$$

uniform wire: radius $r$, length $l$, modulus of rigidity $\eta$ · $C \propto r^4$: halving $r$ makes $T$ FOUR times

## 2 Angular SHM

- $I\alpha = -C\theta$ gives $\alpha = -\omega^2\theta$, $\omega = \sqrt{C/I}$; $\theta = \theta_0 \sin(\omega t + \phi)$.

**Example**

Disc with $I = 0.02\,\mathrm{kg\,m^2}$ on a wire with $C = 0.08\,\mathrm{N\,m\,rad^{-1}}$: $T = 2\pi\sqrt{0.25} = \pi = 3.14\,\mathrm{s}$.

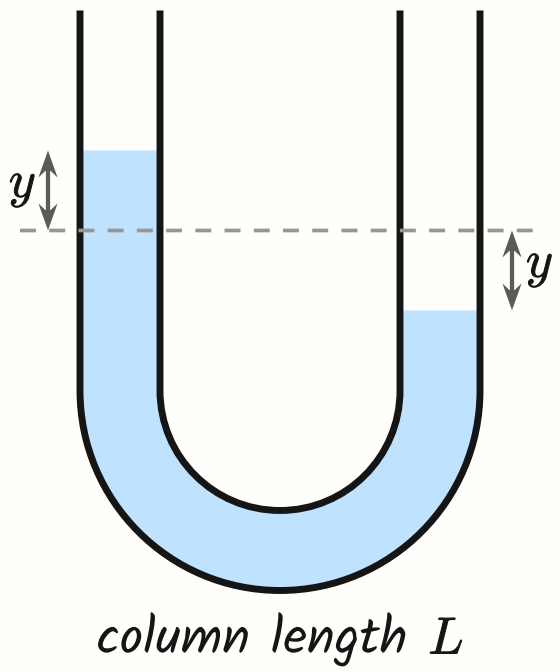
**Golden points**

★ Same wire, two bodies: $\frac{T_1^2}{T_2^2} = \frac{I_1}{I_2}$, so an unknown $I$ is measured by timing.

# U-tube Liquid and Floating Body

*NON-viscous liquid · surface tension neglected · uniform $g$ · small displacement*

## 1 Liquid in a U-tube

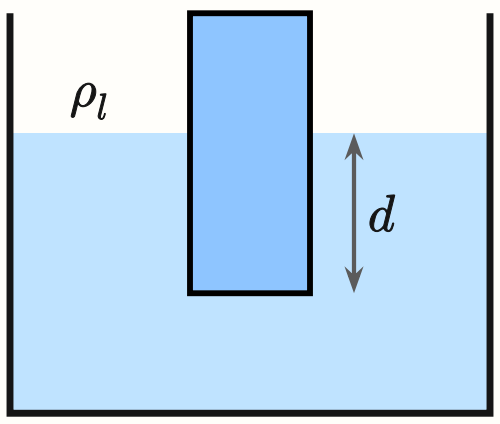



$$T = 2\pi\sqrt{\frac{L}{2g}} = 2\pi\sqrt{\frac{h}{g}}$$

*UNIFORM cross-section, vertical arms · $L$ = total length of the liquid column · $h = L/2$ = rest height in each arm (from the bottom)*

- *Level difference $2y$: force $2y\rho gA$ on mass $\rho AL$, $\omega^2 = \frac{2g}{L}$, any density. $L = 0.4\,\mathrm{m}$: $T = 2\pi\sqrt{0.4/19.6} = 0.898\,\mathrm{s}$.*

## 2 Floating body




$$T = 2\pi\sqrt{\frac{m}{\rho_l A g}} = 2\pi\sqrt{\frac{d}{g}}$$

*vertical cylinder of uniform cross-section $A$, floating UPRIGHT · $d$ = depth under the liquid at equilibrium · drag neglected, never fully submerged or out*

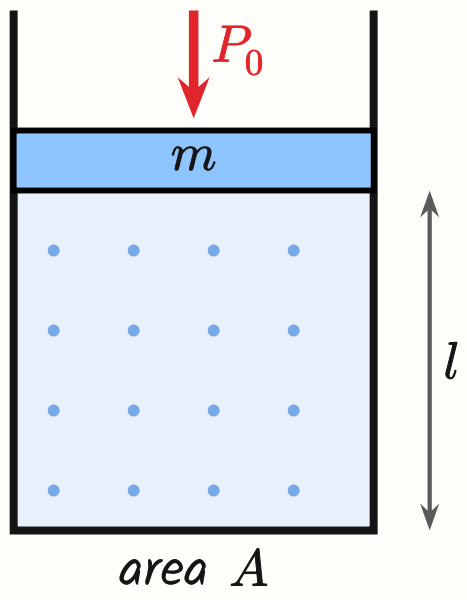
**Example**

$d = 2.45\,\mathrm{cm}$: $T = 2\pi\sqrt{0.0245/9.8} = 2\pi \times 0.05 = 0.314\,\mathrm{s}$.

# Gas Column and Piston

*ideal gas · FRICTIONLESS, gas-tight piston · small displacement* $y \ll l$

## 1 Piston on a gas column



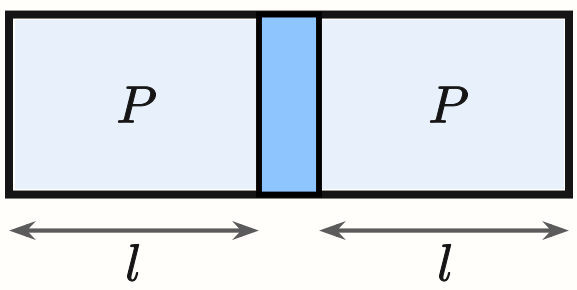

$$T = 2\pi\sqrt{\frac{ml}{PA}} \quad \text{(isothermal)}$$

$$T = 2\pi\sqrt{\frac{ml}{\gamma PA}} \quad \text{(adiabatic)}$$

$P = P_0 + \frac{mg}{A}$ = gas pressure at equilibrium (vertical cylinder) · $l$ = gas column length · $\gamma = C_p/C_v$

- Piston in by $y$: isothermal $\Delta P = \frac{Py}{l}$, so $k = \frac{PA}{l}$; adiabatic $k = \frac{\gamma PA}{l}$.

## 2 Piston between two gas columns




$$T = 2\pi\sqrt{\frac{ml}{2PA}}$$

closed horizontal cylinder, equal columns $l$ at pressure $P$ · isothermal · both sides push back: $k = \frac{2PA}{l}$

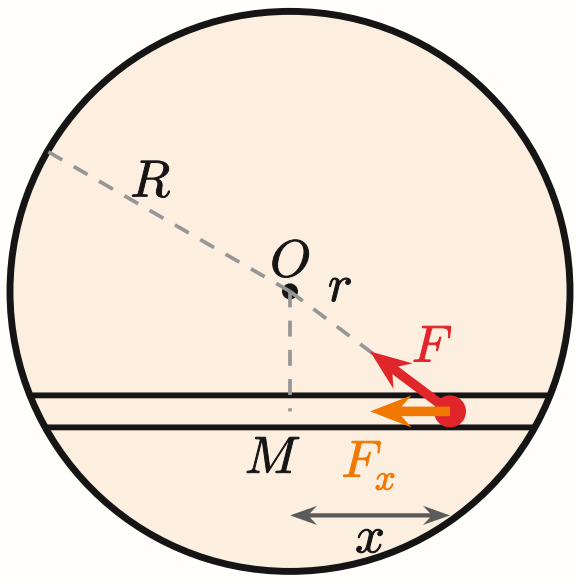
**Example**

$m = 1\,\text{kg}$, $A = 0.01\,\text{m}^2$, $P = 10^5\,\text{Pa}$, $l = 0.1\,\text{m}$ (one column, isothermal): $k = \frac{PA}{l} = 10^4\,\text{N m}^{-1}$, $T = 2\pi\sqrt{10^{-4}} = 0.0628\,\text{s}$.

# Tunnel Through the Earth

**UNIFORM-density earth · smooth tunnel, no air · earth's rotation ignored** · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$

## 1 Force inside the earth




$$F = -\frac{mg}{R}r \;\Rightarrow\; F_x = -\frac{mg}{R}x$$

*inside a uniform sphere only the mass within radius $r$ pulls: $g(r) = g\frac{r}{R}$ · $x$ from the tunnel's mid-point $M$*

## 2 Period, any straight tunnel

$$T = 2\pi\sqrt{\frac{R}{g}} = 84.6\,\mathrm{min}$$

$$v_{centre} = \sqrt{gR} = 7.92\,\mathrm{km\,s^{-1}} \quad \textit{(diametral tunnel)}$$

*same $T$ for EVERY chord (any length) · end to end in $\frac{T}{2} = 42.3\,\mathrm{min}$ · speed at $M$: $\omega\times$ half the chord*

- *$T$ equals the period of a satellite skimming the surface; dropped from the surface, the body starts at an EXTREME, so the amplitude is half the chord.*

**Example**

$\sqrt{R/g} = \sqrt{6.4 \times 10^6/9.8} = 808\,\mathrm{s}$;
$T = 2\pi \times 808 = 5077\,\mathrm{s} = 84.6\,\mathrm{min}$; $\sqrt{gR} = 7.92 \times 10^3\,\mathrm{m\,s^{-1}}$.

# SHM from Potential Energy

*small oscillations about a STABLE equilibrium · one-dimensional motion, conservative force*

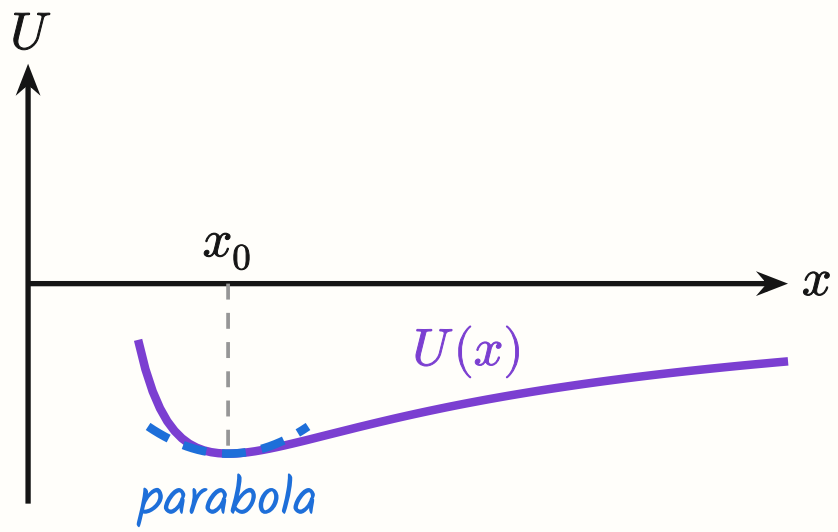
## 1 The rule

$$F = -\frac{dU}{dx} \qquad U'(x_0) = 0,\ U''(x_0) > 0$$

$$\omega^2 = \frac{U''(x_0)}{m} \qquad T = 2\pi\sqrt{\frac{m}{U''(x_0)}}$$

$x_0$ = equilibrium (minimum of $U$) · near $x_0$: $U \approx U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$, so $k_{eff} = U''(x_0)$

## 2 Potential well




- Drawn: $U = \frac{a}{x^2} - \frac{b}{x}$ with $a = 1$, $b = 2$: $x_0 = \frac{2a}{b}$, $U''(x_0) = \frac{b^4}{8a^3}$.
- Maximum of $U$: unstable.
  $U = cx^4$: oscillates, NOT SHM.

## 3 Worked case

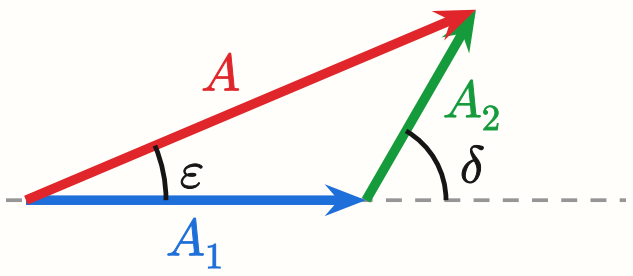
**Example**

$U = (2x^2 - 4x + 5)\,\mathrm{J}$ ($x$ in m), $m = 0.5\,\mathrm{kg}$: $U' = 4x - 4 = 0$ at $x_0 = 1\,\mathrm{m}$; $U'' = 4\,\mathrm{N\,m^{-1}}$; $\omega = \sqrt{4/0.5} = 2.83\,\mathrm{rad\,s^{-1}}$, $T = 2.22\,\mathrm{s}$.

# Superposition: Same Direction

*same direction, SAME angular frequency $\omega$*

## 1 Two SHMs add to one SHM




$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\delta}$$

$$\tan\varepsilon = \frac{A_2\sin\delta}{A_1 + A_2\cos\delta}$$

$x_1 = A_1\sin\omega t$, $x_2 = A_2\sin(\omega t + \delta)$ · *result* $x = A\sin(\omega t + \varepsilon)$, *SAME* $\omega$

- *Equal amplitudes:* $A = 2A_1\cos\frac{\delta}{2}$, $\varepsilon = \frac{\delta}{2}$.
- *Phasors add like vectors.* $\delta = 0$: $A_1 + A_2$. $\delta = \pi$: $|A_1 - A_2|$. $\delta = \frac{\pi}{2}$: $\sqrt{A_1^2 + A_2^2}$.

## 2 Examples

**Example**

$x = 3\sin\omega t + 4\cos\omega t$: $\delta = \frac{\pi}{2}$, $A = 5$, $\tan\varepsilon = \frac{4}{3}$, $\varepsilon = 53°$:
$x = 5\sin(\omega t + 53°)$.
$A_1 = A_2 = 2\,\text{cm}$, $\delta = \frac{2\pi}{3}$: $A = \sqrt{4 + 4 - 4} = 2\,\text{cm}$.

- *Three equal SHMs with phases* $0$, $\frac{2\pi}{3}$, $\frac{4\pi}{3}$: *the phasors close a triangle,* $A = 0$.
- *DIFFERENT* $\omega$: *the sum is not SHM (it is periodic only if* $\omega_1/\omega_2$ *is rational; beats in Waves).*

# Perpendicular SHMs (Lissajous)

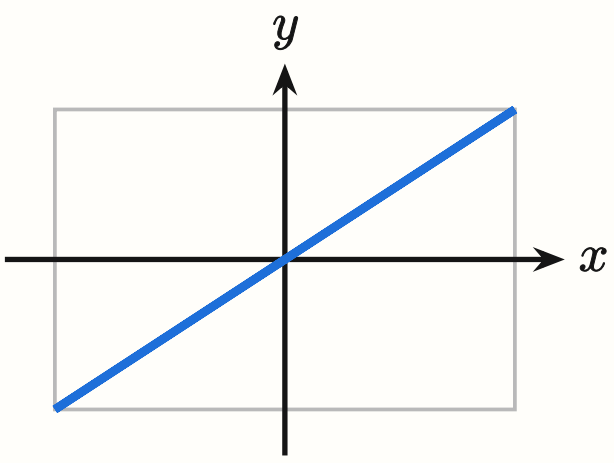
$x = a\sin\omega t$, $y = b\sin(\omega t + \delta)$ · **perpendicular directions, same $\omega$ unless stated**

## 1 Path of the particle

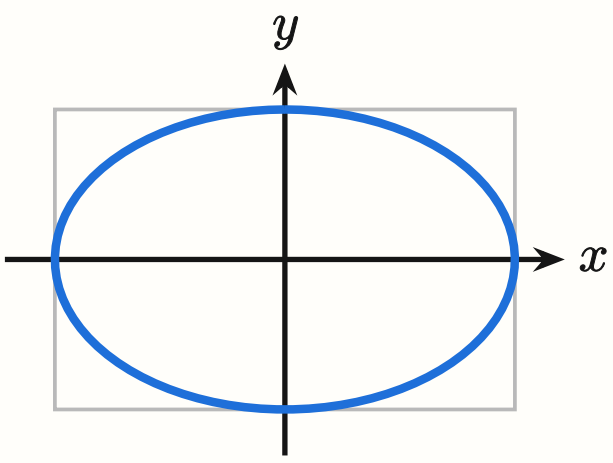
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta$$

eliminate $t$ · in general an ELLIPSE inside the $2a \times 2b$ box

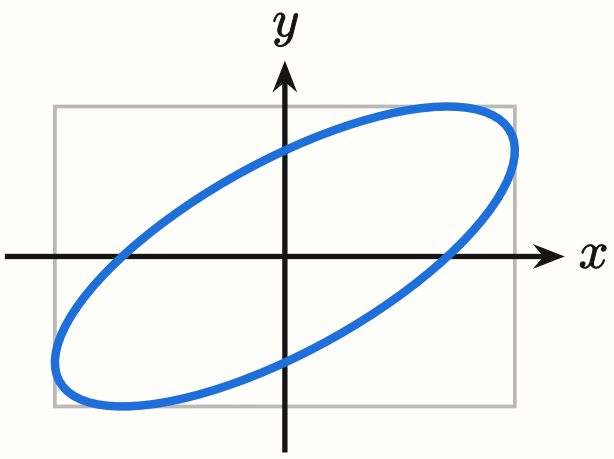
## 2 Standard shapes



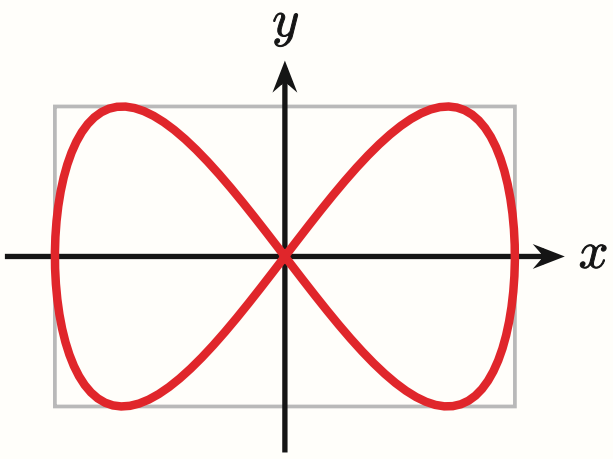

$\delta = 0$: line $y = \frac{b}{a}x$




$\delta = \frac{\pi}{2}$: ellipse




$\delta = \frac{\pi}{4}$: tilted ellipse




$y = b\sin 2\omega t$: figure of eight

- $\delta = \pi$: line $y = -\frac{b}{a}x$. $\delta = \frac{\pi}{2}$, $a = b$: CIRCLE. A line path is an SHM of amplitude $\sqrt{a^2 + b^2}$.

# Damped Oscillations

damping force $-bv$ (proportional to velocity) · LIGHT damping

$b < 2\sqrt{km}$ · $\omega = \sqrt{k/m}$

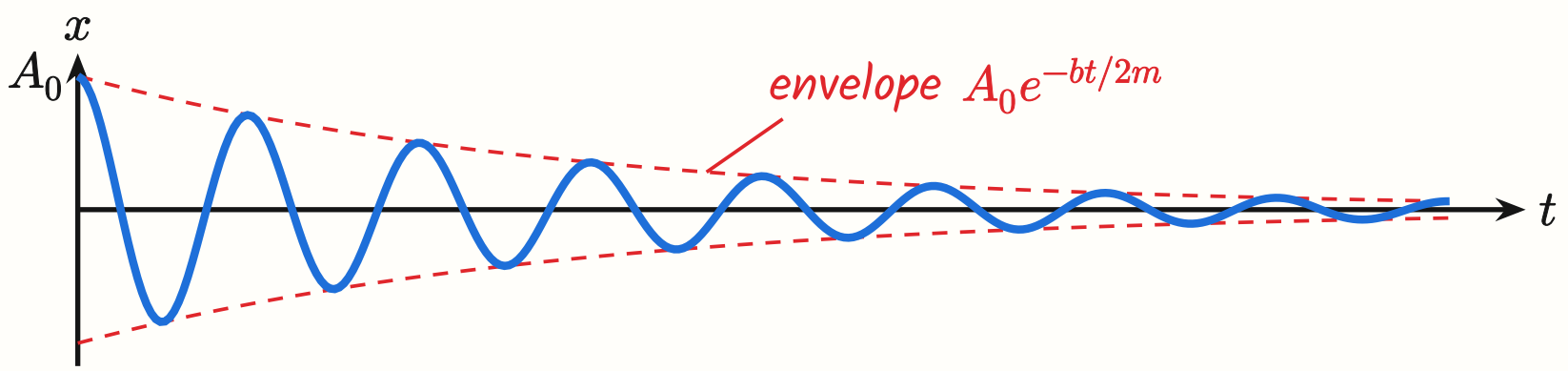
## 1 Equation and solution

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

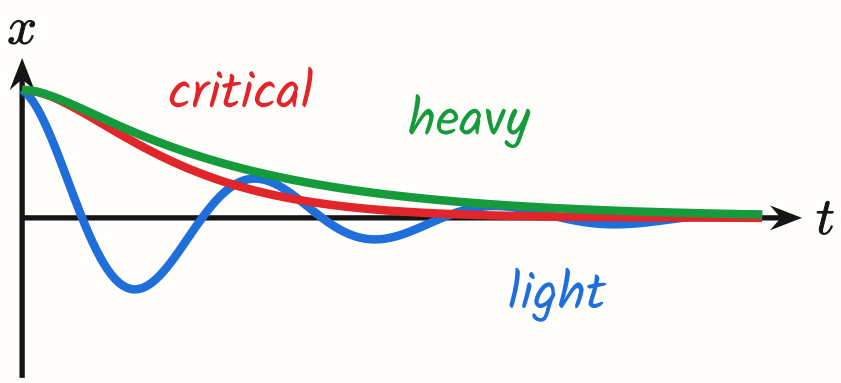
$$x = A_0 e^{-bt/2m}\cos(\omega' t + \phi) \qquad \omega' = \sqrt{\omega^2 - \frac{b^2}{4m^2}}$$

$b$ = damping constant ($\mathrm{kg\,s^{-1}}$) · amplitude $A_0 e^{-bt/2m}$ · $\omega' < \omega$ (slightly slower)

## 2 The graph




## 3 Light, critical, heavy damping




- $b < 2\sqrt{km}$: decaying oscillation.
- $b = 2\sqrt{km}$: critical, FASTEST return, no oscillation.
- $b > 2\sqrt{km}$: overdamped, slow.

# Damping: Energy and Decay

*damping force $-bv$ · LIGHT damping $b < 2\sqrt{km}$ · $t$ from the start, amplitude $A_0$, energy $E_0$*

## 1 Energy and decay times

$$E = E_0 e^{-bt/m}$$

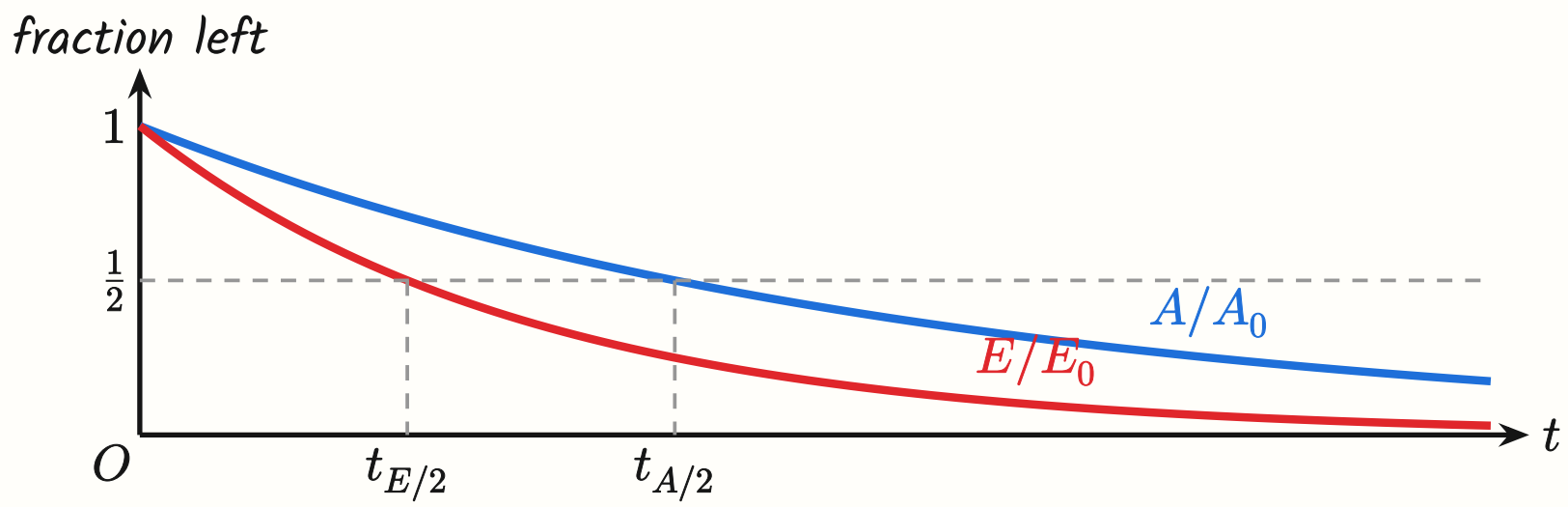
$$t_{A/2} = \frac{2m \ln 2}{b} \qquad t_{E/2} = \frac{m \ln 2}{b}$$

*light damping · energy $\propto$ amplitude$^2$, so it decays TWICE as fast · equal times multiply $A$ by the SAME factor*

**Example**

$m = 0.5\,\text{kg}$, $b = 0.1\,\text{kg s}^{-1}$: $t_{A/2} = \frac{2\times0.5\times0.693}{0.1} = 6.93\,\text{s}$, $t_{E/2} = 3.47\,\text{s}$.

## 2 Amplitude and energy against $t$




## 3 Equal times, equal factors

**Example**

*Amplitude falls to $0.9A_0$ in the first $10\,\text{s}$. After $30\,\text{s}$:* $0.9^3 A_0 = 0.729A_0$; *energy* $0.81^3 E_0 = 0.531E_0$.

# Forced Oscillations and Resonance

**driving force $F_0 \cos \omega_d t$ · damping $-bv$ · $\omega = \sqrt{k/m}$ = natural angular frequency · STEADY state**

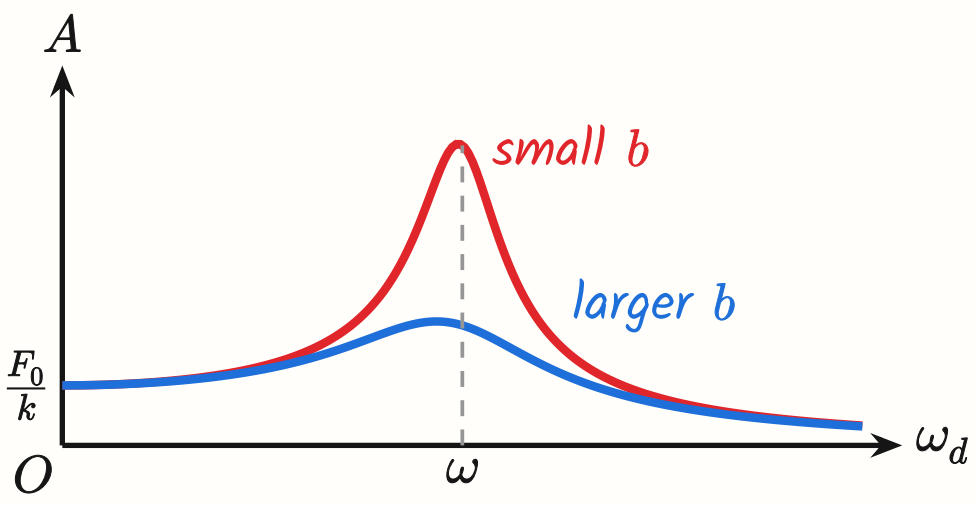
## 1 Steady-state amplitude

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega_d t$$

$$A = \frac{F_0}{\sqrt{m^2(\omega^2 - \omega_d^2)^2 + b^2\omega_d^2}}$$

*after the transient (natural) motion has died out, the body oscillates at the DRIVING frequency $\omega_d$, not at $\omega$*

## 2 Amplitude against driving frequency



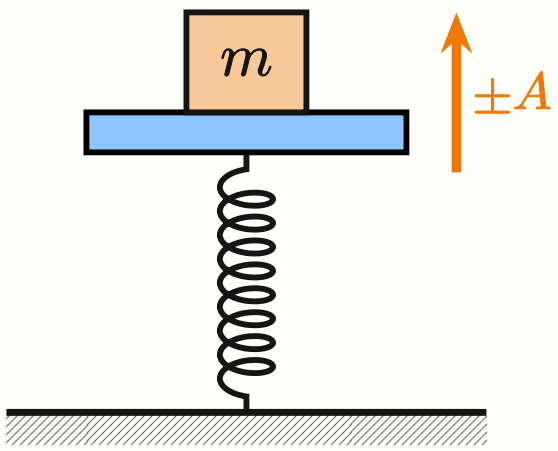

- *RESONANCE: amplitude maximum when $\omega_d \approx \omega$ (light damping).*
- *Smaller $b$: taller, SHARPER peak; $b \to 0$: $A \to \infty$ at $\omega_d = \omega$.*
- *$\omega_d \to 0$: $A \to \frac{F_0}{k}$; $\omega_d \gg \omega$: $A \to 0$.*

- *At $\omega_d = \omega$: $A = \frac{F_0}{b\omega}$ exactly; the true peak, at $\omega_d = \sqrt{\omega^2 - \frac{b^2}{2m^2}}$, is just below $\omega$ (light damping).*
- *Examples: soldiers break step on a bridge; a swing pushed at its own frequency; radio tuning.*

# Special Cases (I)

*ideal springs · uniform* $g = 9.8\,\mathrm{m\,s^{-2}}$ *· small oscillations*

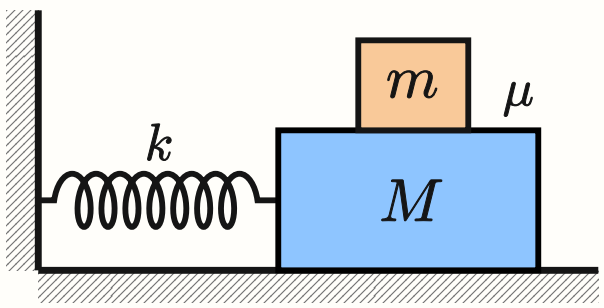
## 1 Block on a vertical spring platform




$$A_{max} = \frac{g}{\omega^2}$$

*block RESTING on a platform in vertical SHM · contact is lost at the TOP if* $A\omega^2 > g$ *(normal force would be negative)*

## 2 Block on a block (friction)




$$A_{max} = \frac{\mu g}{\omega^2} = \frac{\mu g(M+m)}{k}$$

*smooth floor, friction coefficient* $\mu$ *only between the blocks · no slipping if* $A\omega^2 \le \mu g$, $\omega^2 = \frac{k}{M+m}$

## 3 Pendulum clock: temperature and height

$$\frac{\Delta T}{T} = \tfrac{1}{2}\alpha\,\Delta\theta \qquad \frac{\Delta T}{T} = \frac{h}{R}$$

*brass / steel rod, linear expansion coefficient* $\alpha$*, temperature rise* $\Delta\theta$ *· height* $h \ll R$*:* $g$ *falls to* $g(1-\frac{2h}{R})$ *·* $T$ *grows: clock LOSES time*

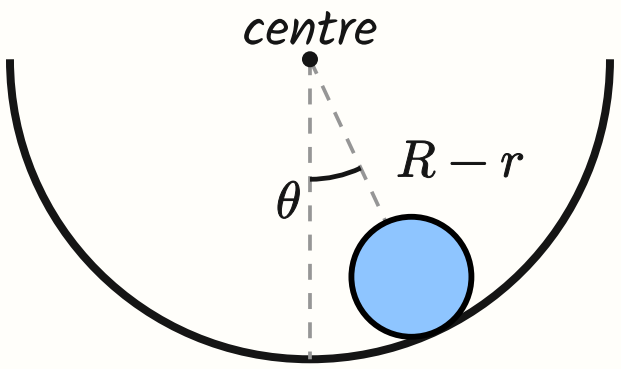
**Example**

$\alpha = 2\times10^{-5}\,^\circ\mathrm{C}^{-1}$, $\Delta\theta = 10^\circ\mathrm{C}$: $\frac{\Delta T}{T} = 10^{-4}$, *loses* $10^{-4}\times 86400 = 8.64\,\mathrm{s}$ *a day. At* $h = 6.4\,\mathrm{km}$: $\frac{h}{R} = 10^{-3}$, *loses* $86.4\,\mathrm{s}$ *a day.*

# Special Cases (2)

*uniform $g$ · small oscillations · ideal springs, smooth floor*

## 1 Ball in a smooth / rough bowl




$$T = 2\pi\sqrt{\frac{R-r}{g}}$$

$$T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$$

*first: SMOOTH bowl (ball slides) · second: solid sphere ROLLING without slipping · bowl radius $R$, ball radius $r$*

## 2 Mass added to an oscillating block

*at the mean:* $A' = A\sqrt{\dfrac{M}{M+m}}$

*at an extreme:* $A' = A$

*block $M$ on a spring $k$, smooth floor · $m$ dropped on gently and sticks (momentum conserved at the mean; at an extreme $v = 0$) · both cases:* $T' = 2\pi\sqrt{\frac{M+m}{k}}$

**Example**

$M = 3\,\text{kg}$, $m = 1\,\text{kg}$, $A = 8\,\text{cm}$, *at the mean:*
$A' = 8\sqrt{0.75} = 6.93\,\text{cm}$; $T$ *grows by* $\sqrt{4/3} = 1.155$ *times.*

# Standard Results

## 1 One-line results

| System | Result | Condition |
|---|---|---|
| SHM | $a = -\omega^2 x$, $v = \omega\sqrt{A^2 - x^2}$ | $x$ from equilibrium |
| Energy | $E = \frac{1}{2}kA^2$ | no damping, PE zero at mean |
| Spring–block | $T = 2\pi\sqrt{m/k}$ | ideal massless spring; horizontal, vertical or incline |
| Series / parallel | $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ / $k_1 + k_2$ | ideal springs |
| Cut spring | $kl$ = constant | uniform spring |
| Massive spring | $2\pi\sqrt{(m + m_s/3)/k}$ | uniform spring, one end fixed |
| Simple pendulum | $T = 2\pi\sqrt{l/g_{eff}}$ | small angle, $\vec{g}_{eff} = \vec{g} - \vec{a}$ |
| Physical | $2\pi\sqrt{I/mgd}$ | rigid body, fixed horizontal axis, small angle |
| Torsional | $2\pi\sqrt{I/C}$ | within the elastic limit |
| U-tube / floating | $2\pi\sqrt{L/2g}$ / $2\pi\sqrt{d/g}$ | non-viscous liquid, uniform cross-section |
| Tunnel | $2\pi\sqrt{R/g}$ | uniform earth, smooth tunnel, any chord |
| From $U(x)$ | $\omega^2 = U''(x_0)/m$ | small oscillations, stable equilibrium |
| Damped | $A_0 e^{-bt/2m}$ | force $-bv$, light damping $b < 2\sqrt{km}$ |

### Summary

$$a = -\omega^2 x \;\bullet\; E = \tfrac{1}{2}kA^2 \;\bullet\; T = 2\pi\sqrt{\frac{m}{k}} \;\bullet\; T = 2\pi\sqrt{\frac{l}{g}}$$

$x$ from equilibrium · no damping · ideal massless spring · pendulum: small angle

# Common Traps

## 1 Exam traps

**Avoid these**

- ✗ Every oscillation is SHM (no: SHM needs a LINEAR restoring force, $a = -\omega^2 x$).
- ✗ $x$ measured from the natural length of a vertical spring (no: from the NEW equilibrium).
- ✗ Pendulum $T$ depends on the mass or on a small amplitude (no; only $l$ and $g_{eff}$).
- ✗ $K$ and $U$ oscillate at $f$ (no: at $2f$, period $\frac{T}{2}$).
- ✗ Average KE over time $= \frac{2}{3}E$ (no: $\frac{E}{2}$; $\frac{2}{3}E$ is the average over POSITION).
- ✗ Cutting a spring in half halves $k$ (no: it DOUBLES $k$).
- ✗ $\tan\phi = \frac{\omega x_0}{v_0}$ fixes $\phi$ (check the quadrant with the signs of $x_0$, $v_0$).
- ✗ Hollow bob slowly leaking water keeps $T$ (no: the centre of mass falls, then rises; $T$ increases, then decreases).
- ✗ A pendulum oscillates in an orbiting satellite (no: $g_{eff} = 0$; a spring–block still gives $2\pi\sqrt{m/k}$).
- ✗ Resonance at $\omega_d = 2\omega$, or forced motion at the natural $\omega$ (no: at $\omega_d \approx \omega$; it moves at $\omega_d$).

**Golden points**

- ★ First ask: where is equilibrium? what is the restoring force for a small $x$? then $\omega^2 = k_{eff}/m_{eff}$.

# Questions: JEE Main Level (I)

*$x$ from equilibrium · ideal massless springs, smooth floor · small angles · $g = 9.8\,\mathrm{m\,s^{-2}}$*

**Q1 · JEE Main level · MCQ**

A particle moves as $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ cm ($t$ in s). Its period and maximum speed: (A) $2\,\mathrm{s}$, $15.7\,\mathrm{cm\,s^{-1}}$ (B) $1\,\mathrm{s}$, $15.7\,\mathrm{cm\,s^{-1}}$ (C) $2\,\mathrm{s}$, $5\,\mathrm{cm\,s^{-1}}$ (D) $\pi\,\mathrm{s}$, $5\,\mathrm{cm\,s^{-1}}$

Solution: $\omega = \pi\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{\omega} = 2\,\mathrm{s}$; $v_{max} = A\omega = 5\pi = 15.7\,\mathrm{cm\,s^{-1}}$.

**Answer: (A)**

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**Q2 · JEE Main level · Numerical**

A $0.5\,\mathrm{kg}$ block on a smooth horizontal floor is attached to an ideal spring of $k = 50\,\mathrm{N\,m^{-1}}$, pulled $4\,\mathrm{cm}$ from equilibrium and released. (i) Maximum speed (in $\mathrm{m\,s^{-1}}$)? (ii) Maximum acceleration (in $\mathrm{m\,s^{-2}}$)?

Solution: $\omega = \sqrt{50/0.5} = 10\,\mathrm{rad\,s^{-1}}$; $v_{max} = A\omega = 0.04 \times 10 = 0.4\,\mathrm{m\,s^{-1}}$; $a_{max} = A\omega^2 = 0.04 \times 100 = 4\,\mathrm{m\,s^{-2}}$.

**Answer: (i) 0.4 (ii) 4**

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**Q3 · JEE Main level · MCQ**

A simple pendulum has period $2\,\mathrm{s}$ on the ground (small oscillations). In a lift accelerating UPWARD at $\frac{g}{3}$ its period is: (A) $1.73\,\mathrm{s}$ (B) $2.31\,\mathrm{s}$ (C) $2\,\mathrm{s}$ (D) $1.5\,\mathrm{s}$

Solution: $g_{eff} = g + \frac{g}{3} = \frac{4g}{3}$; $T = \frac{2}{\sqrt{4/3}} = \sqrt{3} = 1.73\,\mathrm{s}$.

**Answer: (A)**

# Questions: JEE Main Level (2)

*$x$ from equilibrium · ideal massless springs · small angles, frictionless pivot · PE zero at equilibrium · $g = 9.8\,\mathrm{m\,s^{-2}}$*

**Q4 · JEE Main level · Numerical**

A uniform ideal spring of $k = 600\,\mathrm{N\,m^{-1}}$ is cut into three equal parts. Two of the parts, in parallel, hold a $2\,\mathrm{kg}$ block that oscillates vertically. Period (in s)?

Solution: Each part: $3k = 1800\,\mathrm{N\,m^{-1}}$; parallel: $3600\,\mathrm{N\,m^{-1}}$.

$$T = 2\pi\sqrt{\tfrac{2}{3600}} = 0.148\,\mathrm{s}.$$

**Answer:** 0.148

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**Q5 · JEE Main level · MCQ**

A particle performs SHM of amplitude $A$ (no damping, PE zero at equilibrium). Its kinetic energy is THREE times its potential energy at $x =$ (A) $\frac{A}{2}$ (B) $\frac{A}{\sqrt{2}}$ (C) $\frac{A}{3}$ (D) $\frac{A}{\sqrt{3}}$

Solution: $\frac{1}{2}k(A^2 - x^2) = 3 \times \frac{1}{2}kx^2$ gives $A^2 = 4x^2$, $x = \frac{A}{2}$.

**Answer: (A)**

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**Q6 · JEE Main level · Numerical**

A uniform rod of length $0.6\,\mathrm{m}$ swings with small amplitude about a fixed horizontal axis through one end (frictionless pivot). Period (in s)?

Solution: $I = \frac{mL^2}{3}$, $d = \frac{L}{2}$; $T = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{1.2}{29.4}} = 1.27\,\mathrm{s}.$

**Answer:** 1.27

# Questions: JEE Main Level (3)

**damping force $-bv$ · NON-viscous liquid, uniform U-tube · $g = 9.8\,\mathrm{m\,s^{-2}}$**

**Q7 · JEE Main level · MCQ**

A $0.25\,\mathrm{kg}$ block on a spring of $k = 25\,\mathrm{N\,m^{-1}}$ has damping constant $b = 0.05\,\mathrm{kg\,s^{-1}}$ (light damping). Time for its amplitude to halve:
(A) $6.93\,\mathrm{s}$ (B) $3.47\,\mathrm{s}$ (C) $13.9\,\mathrm{s}$ (D) $10\,\mathrm{s}$

Solution: $b < 2\sqrt{km} = 5\,\mathrm{kg\,s^{-1}}$. $e^{-bt/2m} = \frac{1}{2}$:

$t = \frac{2m\ln 2}{b} = \frac{0.5\times 0.693}{0.05} = 6.93\,\mathrm{s}.$

**Answer: (A)**

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**Q8 · JEE Main level · Numerical**

Two SHMs along the same line with the same frequency have amplitudes $3\,\mathrm{cm}$ and $4\,\mathrm{cm}$ and a phase difference $\frac{\pi}{3}$. Resultant amplitude (in cm)?

Solution: $A = \sqrt{9 + 16 + 2\times 3\times 4\times \frac{1}{2}} = \sqrt{37} = 6.08\,\mathrm{cm}.$

**Answer:** 6.08

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**Q9 · JEE Main level · Numerical**

A U-tube of uniform cross-section holds a liquid column of total length $0.98\,\mathrm{m}$. The liquid is displaced slightly and released. Period (in s)?

Solution: $T = 2\pi\sqrt{\frac{L}{2g}} = 2\pi\sqrt{\frac{0.98}{19.6}} = 2\pi\sqrt{0.05} = 1.40\,\mathrm{s}$, whatever the liquid's density.

**Answer:** 1.40

# Questions: JEE Advanced Level (I)

*one-dimensional conservative motion · smooth floor · ideal spring · $g = 9.8\,\mathrm{m\,s^{-2}}$*

**A1 · JEE Advanced level · Multi-correct**

A $0.1\,\mathrm{kg}$ particle moves along $x$ in the potential $U = (5x^2 - 10x + 2)\,\mathrm{J}$ ($x$ in m) with total mechanical energy $2\,\mathrm{J}$. Then: (A) equilibrium is at $x = 1\,\mathrm{m}$ (B) the period is $\frac{\pi}{5}\,\mathrm{s}$ (C) the amplitude is $1\,\mathrm{m}$ (D) the maximum speed is $5\,\mathrm{m\,s^{-1}}$

Solution: $U' = 10x - 10 = 0$ at $x_0 = 1\,\mathrm{m}$; $U'' = 10\,\mathrm{N\,m^{-1}}$, $\omega = \sqrt{10/0.1} = 10\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{10} = \frac{\pi}{5}\,\mathrm{s}$. $U(x_0) = -3\,\mathrm{J}$, so $\frac{1}{2} \times 10A^2 = 2 - (-3) = 5$: $A = 1\,\mathrm{m}$. $K_{max} = 5\,\mathrm{J}$: $v_{max} = \sqrt{2 \times 5/0.1} = 10\,\mathrm{m\,s^{-1}}$, not 5.

**Answer: (A), (B), (C)**

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**A2 · JEE Advanced level · Numerical**

A $2\,\mathrm{kg}$ block on a smooth floor is attached to a wall by an ideal spring of $k = 200\,\mathrm{N\,m^{-1}}$. A $0.5\,\mathrm{kg}$ block rests on it; the coefficient of static friction between the blocks is $0.6$. Largest amplitude (in cm) for which the top block does not slip?

Solution: Together: $\omega^2 = \frac{k}{M+m} = \frac{200}{2.5} = 80\,\mathrm{s^{-2}}$. The top block needs $m\omega^2 A \le \mu m g$: $A_{max} = \frac{\mu g}{\omega^2} = \frac{0.6 \times 9.8}{80} = 0.0735\,\mathrm{m} = 7.35\,\mathrm{cm}$.

**Answer: 7.35**

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**Golden points**

★ Friction on the top block supplies ITS share of the SHM force: largest at the extremes, $m\omega^2 A$.

# Questions: JEE Advanced Level (2)

*small angles · uniform-density earth, smooth tunnel, no air, rotation ignored · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$*

**A3 · JEE Advanced level · Matching**

A simple pendulum has period $T_0$ at rest. List I: (P) in a lift accelerating up at $\frac{g}{2}$ (Q) in a lift going down at constant speed (R) in a cart accelerating horizontally at $g$ (S) bob of density twice that of a NON-viscous liquid it is fully immersed in (drag neglected). List II: (1) $T_0$ (2) $\sqrt{2}\,T_0$ (3) $0.816\,T_0$ (4) $0.841\,T_0$

Solution: $T \propto g_{eff}^{-1/2}$. (P) $1.5g$: $\sqrt{2/3} = 0.816$. (Q) $g$: $T_0$. (R) $\sqrt{2}g$: $2^{-1/4} = 0.841$. (S) $g(1-\frac{1}{2}) = \frac{g}{2}$: $\sqrt{2}\,T_0$.

**Answer: P-3, Q-1, R-4, S-2**

**A4 · JEE Advanced level · Numerical**

A straight smooth tunnel is dug along a chord whose distance from the earth's centre is $\frac{R}{2}$. A ball is released from rest at one end. (i) Time (in min) to reach the other end? (ii) Maximum speed (in $\mathrm{km\,s^{-1}}$)?

Solution: $T = 2\pi\sqrt{R/g} = 84.6\,\mathrm{min}$; end to end $= \frac{T}{2} = 42.3\,\mathrm{min}$.

Amplitude = half the chord $= \sqrt{R^2 - \frac{R^2}{4}} = \frac{\sqrt{3}}{2}R$;

$v_{max} = \omega A = \frac{\sqrt{3}}{2}\sqrt{gR} = 0.866 \times 7.92 = 6.86\,\mathrm{km\,s^{-1}}$.

**Answer: (i) 42.3 (ii) 6.86**

**Golden points**

★ The chord changes the AMPLITUDE (and so the speed), never the period.

# Questions: JEE Advanced Level (3)

**driving force $F_0\cos\omega_d t$, damping $-bv$ · rigid body, fixed horizontal axis, small angle · $g = 9.8\,\mathrm{m\,s^{-2}}$**

**A5 · JEE Advanced level · Multi-correct**

A $0.1\,\mathrm{kg}$ block on an ideal spring of $k = 10\,\mathrm{N\,m^{-1}}$ (damping constant $b = 0.2\,\mathrm{kg\,s^{-1}}$) is driven by $F_0\cos\omega_d t$ with $F_0 = 0.5\,\mathrm{N}$. In the steady state: (A) the natural angular frequency is $10\,\mathrm{rad\,s^{-1}}$ (B) at $\omega_d = 10\,\mathrm{rad\,s^{-1}}$ the amplitude is $0.25\,\mathrm{m}$ (C) the block oscillates at $\omega_d$ (D) as $\omega_d \to 0$ the amplitude tends to $0.5\,\mathrm{m}$

Solution: $\omega = \sqrt{10/0.1} = 10\,\mathrm{rad\,s^{-1}}$; $b < 2\sqrt{km} = 2\,\mathrm{kg\,s^{-1}}$. At $\omega_d = \omega$: $A = \frac{F_0}{b\omega} = \frac{0.5}{0.2\times 10} = 0.25\,\mathrm{m}$. As $\omega_d \to 0$: $A \to \frac{F_0}{k} = 0.05\,\mathrm{m}$.

**Answer: (A), (B), (C)**

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**A6 · JEE Advanced level · Numerical**

A uniform disc of radius $0.2\,\mathrm{m}$ oscillates in its own vertical plane about a horizontal axis $\perp$ the disc through a point at distance $d$ from its centre. (i) $d$ (in cm) for the SMALLEST period? (ii) That period (in s)?

Solution: $k^2 = \frac{R^2}{2}$, $T = 2\pi\sqrt{\frac{k^2+d^2}{gd}}$ is least at $d = k = \frac{R}{\sqrt{2}} = 0.141\,\mathrm{m}$.

$T_{min} = 2\pi\sqrt{\frac{2k}{g}} = 2\pi\sqrt{\frac{0.283}{9.8}} = 1.07\,\mathrm{s}$.

**Answer: (i) 14.1 (ii) 1.07**

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**Golden points**

- ★ Resonance amplitude $\frac{F_0}{b\omega}$ is set by DAMPING alone; far below resonance it is the static $\frac{F_0}{k}$.
- ★ Physical pendulum: $T(d)$ has a minimum at $d = k$; two values of $d$ give every larger period.